The High-Redshift Accretion Atlas

A Tour of the Final Catalogue and Paper-Ready Results

Repository status document - 2 September 2026

Executive summary

Versions identify nested scientific datasets rather than software releases. v1 is the complete analysis of the original 23-object JADES broad-line AGN catalogue; v2 applies the same analysis to 112 comparable broad-line AGN; and v3 is the final heterogeneous atlas within the 27 August 2026 source-family review cutoff. It contains 234 measurements, 219 physical objects, and 218 host systems.

All three datasets use the same corrected identity rules, cosmology, growth model, uncertainty propagation, comparison policy, and visual grammar. In v3, 196 objects support numerical growth inference and 23 remain visible as explicit no-inference cases. Results are diagnostic comparisons under stated assumptions, not evidence for a unique seed or accretion history.

Canonical dataset scopes

Version	Scope	Measurements	Objects	Hosts	Numeric
v1	Original JADES BLAGN	23	23	23	23
v2	Expanded comparable BLAGN	119	112	111	112
v3	Heterogeneous admitted atlas	234	219	218	196

Canonical catalogues live under `data/processed/<version>/`; identity products under `data/crossmatch/<version>/`; and numerical and visual products under `results/<version>/`.

The final catalogue

The landscape separates growth-eligible preferred objects from retained catalogue-only evidence records. Alternate literature measurements are preserved, but one documented preferred measurement drives each object-level result.

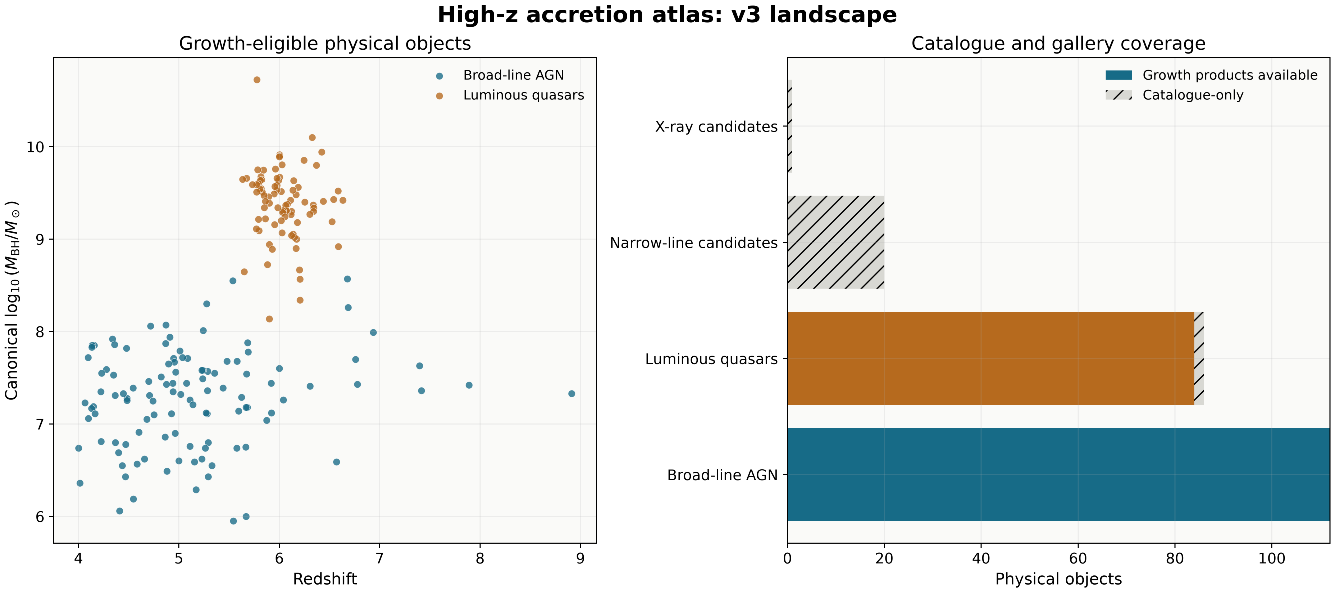

Figure 1. *Final v3 catalogue landscape. Heterogeneous classes remain visually distinct and unsupported masses are not invented.*

Growth model and complete object coverage

The atlas evaluates a Dayal-style exponential black-hole growth relation with cosmic time from a flat Planck-style cosmology. The baseline uses seed redshift 30, radiative efficiency 0.1, and no merger boost. It solves both for the lifetime-average Eddington ratio at fixed seed mass and for seed mass at fixed accretion history.

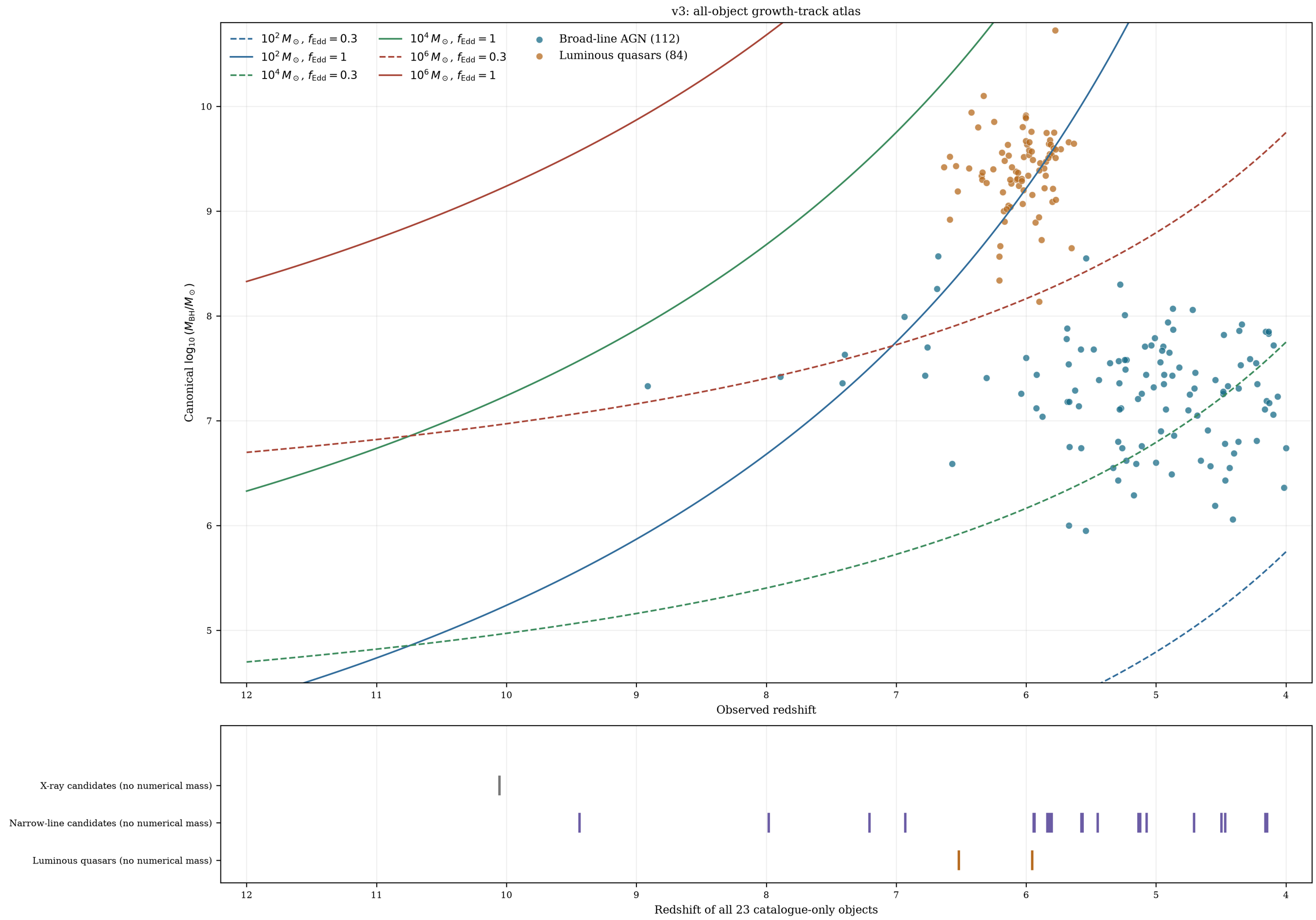

Figure 2. *All 196 supported masses appear against reference tracks; the lower panel retains the redshifts and classes of all 23 no-inference objects.*

Class-aware growth pressure

Global ranks are navigation aids only. Scientific comparisons are retained within object class and mass-comparability group, and pooled demographic inference is explicitly disallowed.

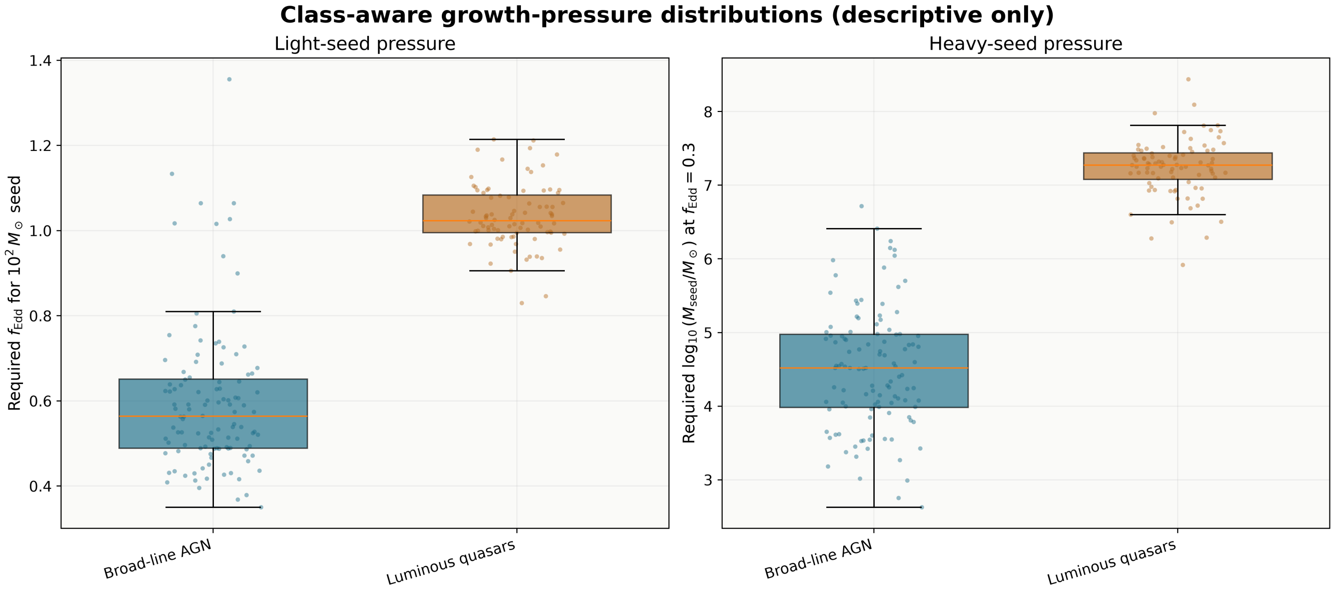

Figure 3. *Required accretion and seed-mass pressure by object class. The distributions are descriptive because the source selections are heterogeneous.*

Uncertainty and duty-cycle diagnostics

Every eligible measurement and object has 10,000 deterministic Monte Carlo draws from reported asymmetric black-hole mass uncertainties. Statistical errors remain separate from source-specific virial systematics. Two-state histories retain duty cycles above one as evidence that a fixed burst scenario is insufficient.

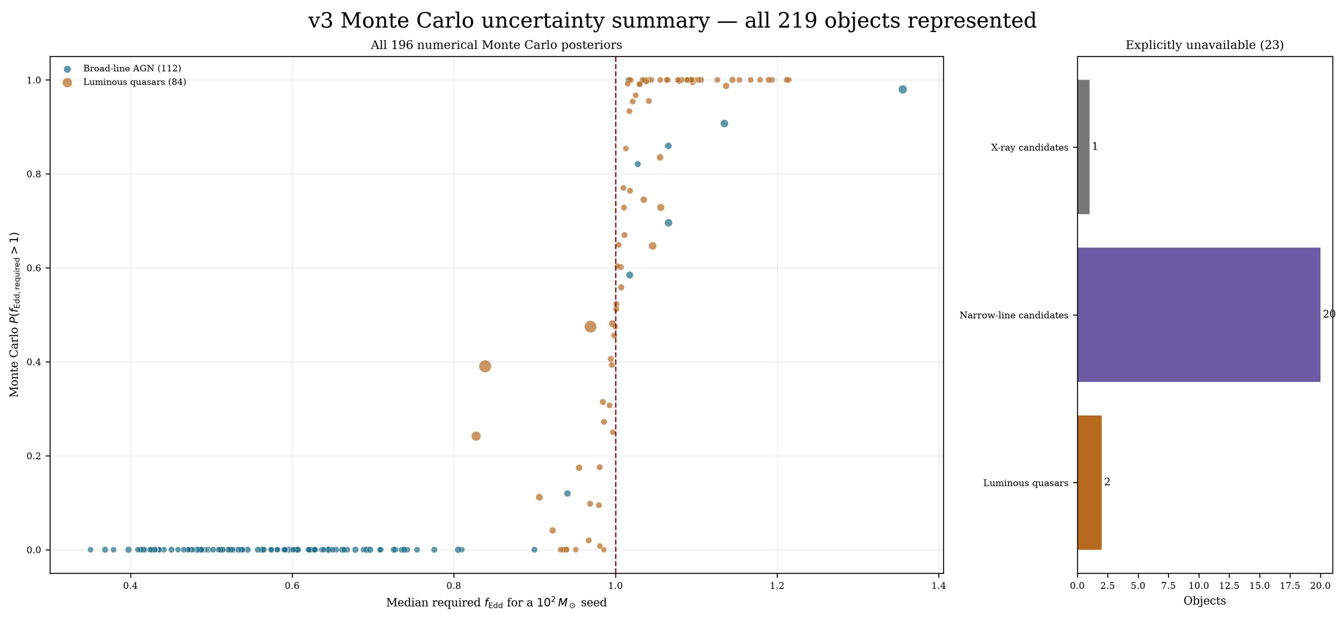

Figure 4. *Monte Carlo pressure summary for all 196 numerical objects, with an explicit accounting of the 23 unavailable cases.*

Compatibility across assumption families

Compatibility is evaluated object by object across four seed families, three spin cases, two merger cases, and three lifetime-average Eddington ratios. Unsupported objects have unavailable values rather than false incompatibilities.

Figure 5. *Class-specific compatibility fractions across seed, spin, merger, and accretion assumptions.*

From catalogue scale to individual objects

Each of the 219 v3 objects has a parameter-map sheet, a growth-track panel, and a seed-redshift panel. The 196 supported objects receive numerical products; the remaining 23 receive status panels explaining why no inference is made.

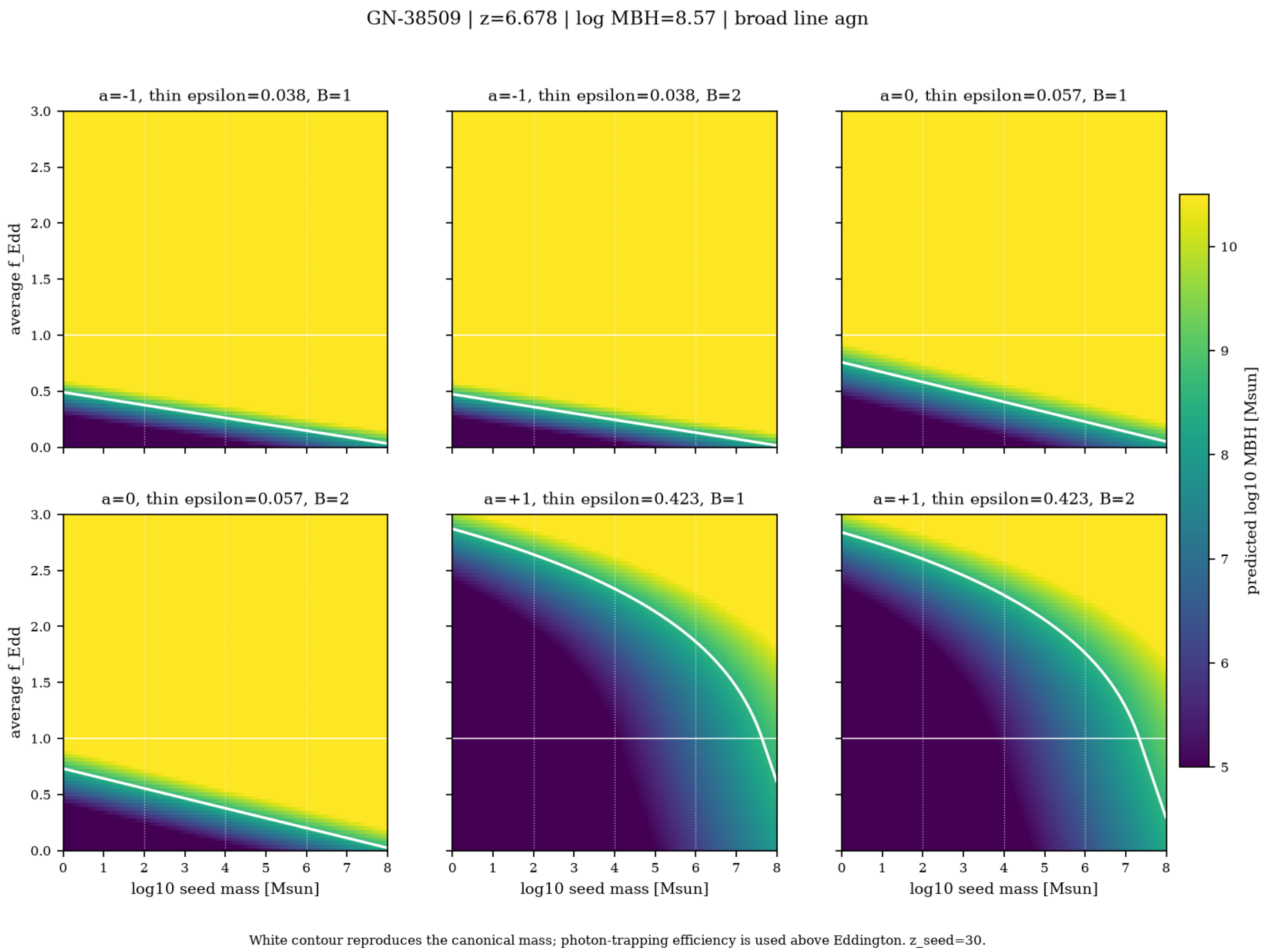

Figure 6. *GN-38509 parameter-map sheet. Six spin/merger combinations show which seed and accretion choices reproduce the preferred mass.*

Measurement choice and provenance

Thirteen eligible alternate measurements are compared with preferred rows. Source keys, table locations, publication versions, archive records, uncertainties, identity decisions, and caveats remain available beside derived quantities. The follow-up matrix retains all 219 objects (196 ranked; 23 explicitly unranked), and the source caveat table covers all 11 admitted families.

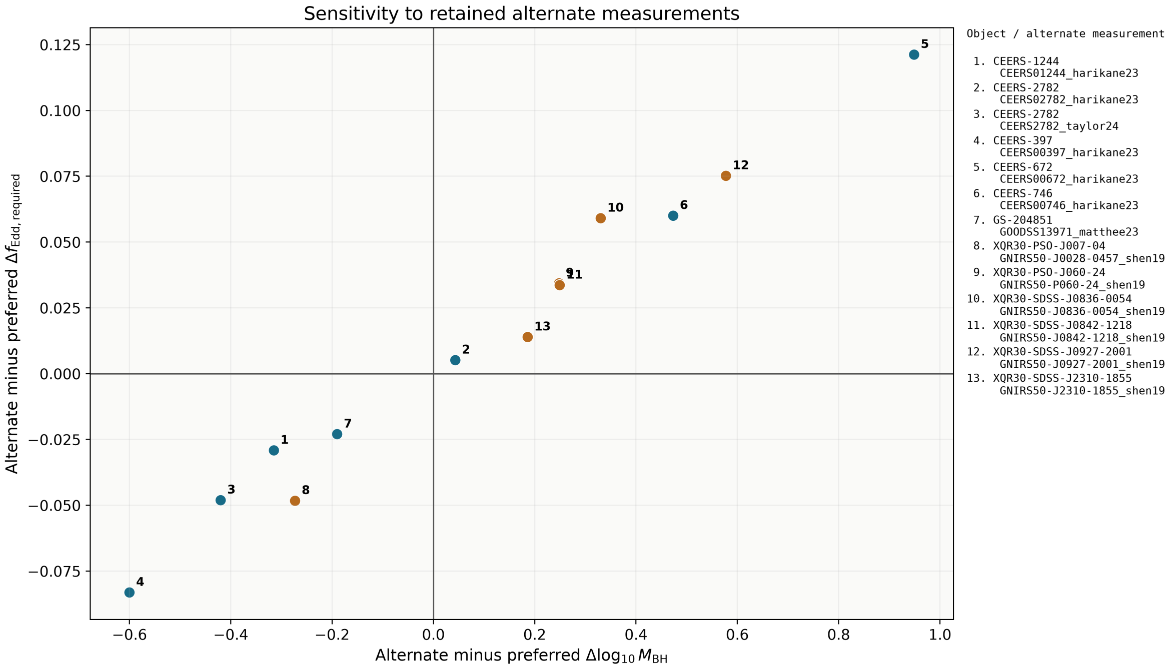

Figure 7. *Sensitivity of mass and required accretion to retained alternate measurements.*

Repository workflow and verification

Public workflows are the ordered notebooks under `scripts/`. They call testable Python implementation under `src/internal/`. Only the source-admission builders required for exact reconstruction retain historical names under `src/internal/compatibility/`. Obsolete release-era data, results, manifests, and documentation were removed from the public tree and remain recoverable from Git history.

Order	Notebook	Purpose
1	00_process_catalogues.ipynb	Materialize v1/v2/v3 catalogues
2	01_generate_science.ipynb	Rankings, uncertainty, duty cycles
3	02_generate_figures.ipynb	Paper summary figures
4	03_generate_atlas.ipynb	Atlases, result inventory, manifests
5	04_verify.ipynb	Provenance, manifests, exact reproduction, tests

The final verification gate checks strict v1 < v2 < v3 measurement membership, catalogue and analysis cardinalities, the result inventory, canonical SHA-256 manifests, exact in-memory CSV reproduction, 10,000-draw products, compatibility coverage, image resolution, and three per-object panels for every catalogue object.

Literature boundary and next dataset

The canonical source-family review cutoff is 27 August 2026. v3 is final within its declared admitted-source scope, not an evergreen exhaustive census. Relevant sources that require new adapters or comparability policy are listed in `docs/current/literature-scope.md` and will create a new dataset version rather than mutate v3.

Current status

The migration is structurally complete: v1/v2/v3 are the public datasets; canonical data, tables, figures, galleries, and manifests are retained; implementation is Python under src; user workflows are notebooks; contribution history and this status document are versioned; and the required historical source-admission bridge is isolated as internal compatibility code.